

Abdelrahman Saeed

Artificial Intelligence Engineer

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SUMMARY

Artificial Intelligence Engineer with a strong software engineering background in Python, C#, and C++. Experienced in building end-to-end machine learning pipelines including data cleaning, preprocessing, feature engineering, model training, evaluation, optimization, and deployment. Improved decision accuracy by 30% and increased model performance by 25–40% across classification and regression tasks. Integrated machine learning solutions with scalable backend systems, improving production readiness by 35% and enabling reliable, maintainable deployment in real-world applications.

EDUCATION

Bachelor's degree in computer science and IT | Al-Ahram Canadian University | 2022 – 2026 **GPA: 3.68 / 4**

WORK EXPERIENCE

AI Engineer — Logan Travel

Jan 8, 2026, to Jul 1, 2026

- Designed and independently built an end-to-end AI Agent that transformed manual document analysis, summarization, and reporting workflows into automated intelligent processes, supporting Excel, PDF, DOCX, CSV, and TXT files.
- Engineered a specialized Arabic Excel analysis pipeline capable of processing complex human-designed workbooks with merged cells, irregular headers, manually structured tables, empty rows, and multiple tables across 24+ sheets, covering sales, accounting, employee, inventory, and related business data.
- Built an intelligent query-routing and execution system that classifies user intent and dynamically routes questions between RAG-based retrieval and structured Excel analysis; generated validated JSON queries through LLMs and executed aggregation and mathematical operations using Pandas.
- Developed a hybrid RAG pipeline using Unstructured, title-based chunking, Sentence Transformers, Chroma, BM25, Ensemble Retrieval, RRF, and Cohere reranking, with Qwen3 and Gemini 2.5 Flash selected according to the processing stage.
- Reduced manual Excel report-generation time from 30–45 minutes to approximately 5 minutes, while maintaining accurate results reviewed by management, and integrated the AI subsystem into the company's production system.

Web Developer / Bot Developer

Nov 11, 2024, to Mar 22, 2025

WhatsApp Customer Service Bot

- Developed a WhatsApp bot integrated with Meta's Business API to streamline customer service interactions.
- Built a dynamic admin panel enabling non-technical users to manage categories, issues, and solution links without modifying the codebase.
- Designed and implemented RESTful APIs using Flask.
- Engineered the backend with SQLite for lightweight and efficient data management.
- Developed a responsive front end using HTML, CSS, and JavaScript to enhance usability for admin users.
- Deployed the complete system on Railway cloud hosting, ensuring reliable availability, environment configuration, and real-world usage.

TECHNICAL PROJECTS

Deep Learning – Smart Stock ERP System

Graduation Project – 2026, Grade A

- Designed and trained a GRU-based neural network to forecast 30-day sales revenue from historical ERP transaction data, using a sliding 30-day input window.
- Built the full data pipeline: extracted invoices from MySQL, aggregated daily sales, engineered calendar features (day-of-week, month, event flags), and applied log-scaling/normalization for stable training.

- Implemented autoregressive multi-step forecasting, feeding each day's prediction back into the model to generate a full 30-day forecast.
- Built a retraining pipeline with early stopping, plus automated MAPE-based accuracy monitoring that flags unreliable forecasts and triggers retraining.
- Integrated the model into a Node.js/Express backend via process orchestration, exposing forecasts through REST APIs secured with JWT role-based access and transaction-safe inventory logic.
- Independently designed and built the entire backend of the ERP system — covering employee management, inventory, purchasing, outbound/inbound workflows, invoicing, and dashboards — with transaction-safe stock logic (row locking) and JWT-based role access control across all modules.

Machine Learning – E-Commerce Sales Prediction

Personal Project – 2025

- Performed in-depth exploration data analysis and visualization on a 500,000-transaction UK e-commerce dataset, uncovering seasonal sales trends like top-performing products.
- Performed feature engineering, including cleaning (missing values, outliers, duplicates), temporal feature extraction, target variable creation, and feature scaling to prepare high-quality data for modeling.
- Built, trained, and compared six regression models to predict total sales; evaluated performance using MSE, MAE, and R² score, conducted hyperparameter tuning, and identified the best-performing model.

Machine Learning – Cardiovascular Disease Classification

Personal Project – 2025

- Built and deployed an end-to-end machine learning pipeline on 70,000+ medical records, performing data cleaning, enforcing medical constraints, and engineering clinically meaningful features (BMI, pulse pressure).
- Trained, tuned, and compared multiple classification models (Logistic Regression, KNN, SVM, Random Forest, Gradient Boosting, XGBoost) using cross-validation and GridSearchCV, identifying XGBoost as the best-performing model based on ROC-AUC, precision, recall, and F1-score.
- Saved the final optimized model using the pickle library and deployed it as an interactive Streamlit web application, enabling real-time cardiovascular disease prediction.

Deep Learning – Brain Tumor Classification

Personal Project – 2025

- Performed data exploration and visualization of MRI images to identify visual differences between brain gliomas and meningiomas, and to analyze image size and pixel density.
- Applied intensive data amplification techniques to improve model robustness and generalizability.
- Applied a custom CNN, ResNet50, and VGG16 using transfer learning, comparing their performance through accuracy, recall, F1 score, confusion matrices, fine-tuning analysis, and training curves.

CERTIFICATIONS & COURSES

- Artificial Intelligence | ALX Academy | 2024
- Machine Learning | Mindset Training | 2025
- Deep Learning | Mindset Training | 2025
- AI & Machine Learning Foundations | Sprints | 2025
- HCIA-Big Data | October 6 University | 2025

SKILLS

Technical skills: Programming Languages: C++, Python, C#, SQL, Excel, Dart | Frameworks & Tools: Sklearn, Plotly, NumPy, Pandas, TensorFlow, joblib, Matplotlib, Keras, Streamlit, Git, GitHub, Flask, Django, Node.js, FastAPI | Databases: SQLite, MySQL, PostgreSQL, Microsoft SQL Server | Architecture: RESTful API Design, CRUD, REST inference endpoints.

Soft skills: Agile | Scrum | Strong problem-solving | Effective communication | Deep critical thinking | Exceptional patience in research and information gathering | Research & Information Gathering | Extreme Programming.

LANGUAGES

- Arabic (Native)
- English (Intermediate – B2)